COMP 3920 Database Systems 1

Hosted Cloudinary Images

Introduction:

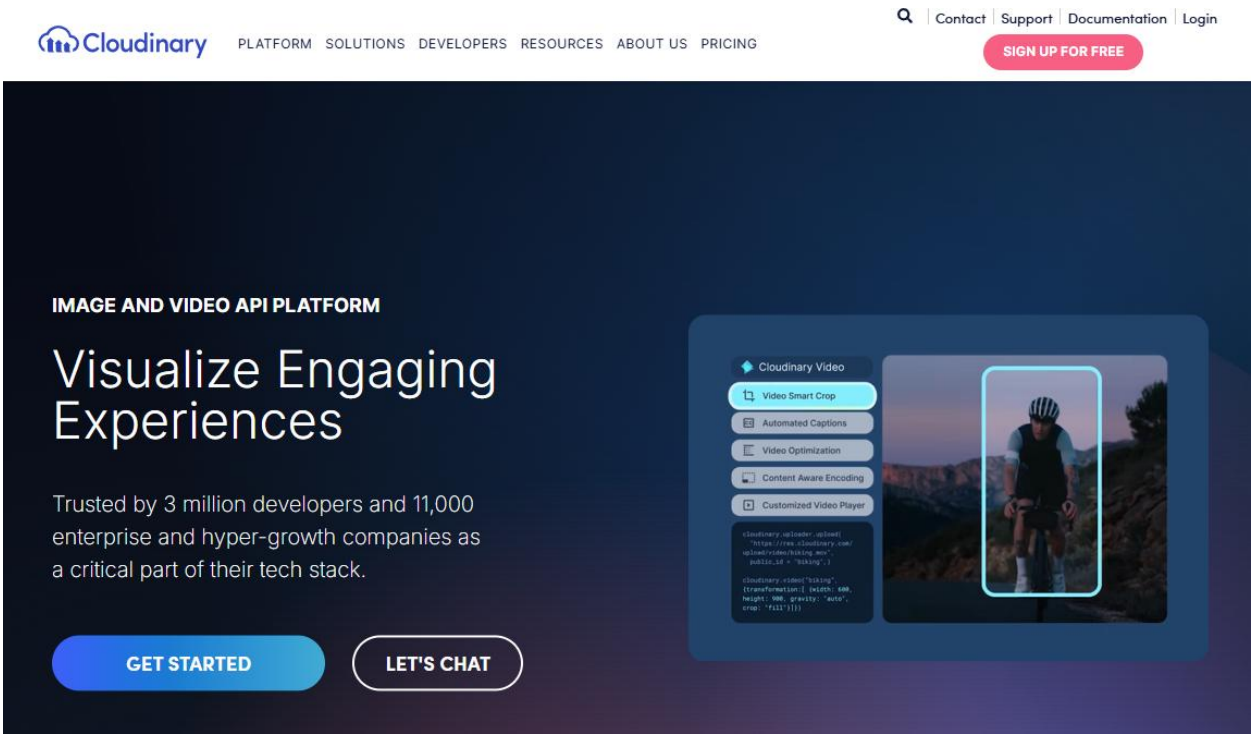
The goal of this document is to serve as a walkthrough for working with images in a database.

Storing images in a database can create some challenges. Although it is technically possible to store the images directly into the database (using a BLOB datatype), it isn't recommended if you have a lot of files because this tends to increase the size of your database very quickly. A better approach is to store the images on a Content Delivery Network (CDN) and then simply store a link to the file on the CDN in the database.

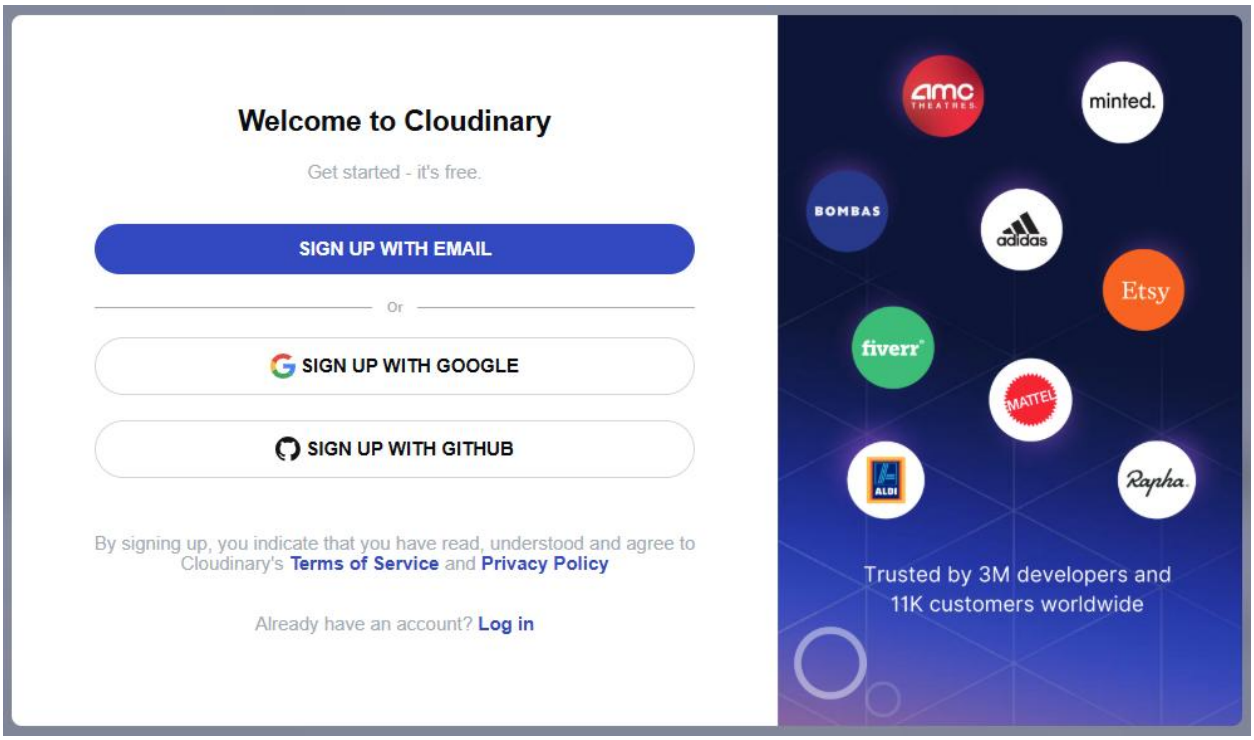
In this lab we'll be using Cloudinary as a CDN service. There are several other CDNs, for example Amazon S3, Google Cloud Storage, Azure Blob Storage, etc. Cloudinary is a cloud-based service that stores your images and videos and automatically optimizes them for fast loading on any device. It also lets you resize, crop, format, and enhance media on the fly using simple URLs, so you don’t have to manage multiple versions yourself.

Step 1: Get a Cloudinary Account.

Goto cloudinary.com

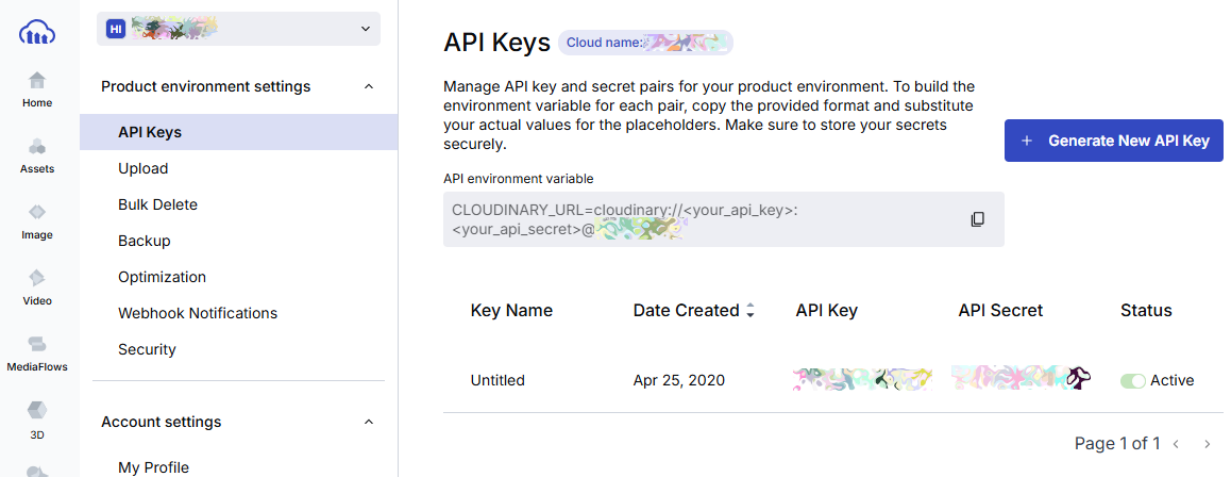


Sign up for a new account (if you don't already have one).



Take note of your:

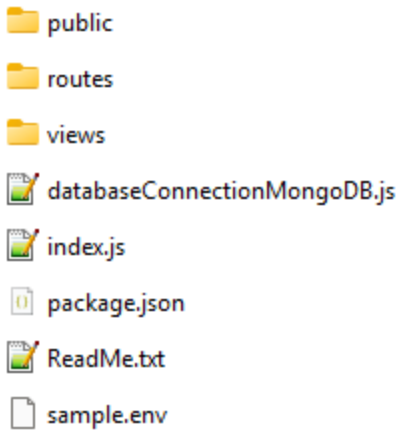
1. Cloud Name:
2. API Key:
3. API Secret:



Step 2: Create a GitHub Repo.

Download the `Cloudinary Example.zip` zip file from D2L (learn.bcit.ca).

Extract all files from the zip file into a folder. Your folder should look like this:



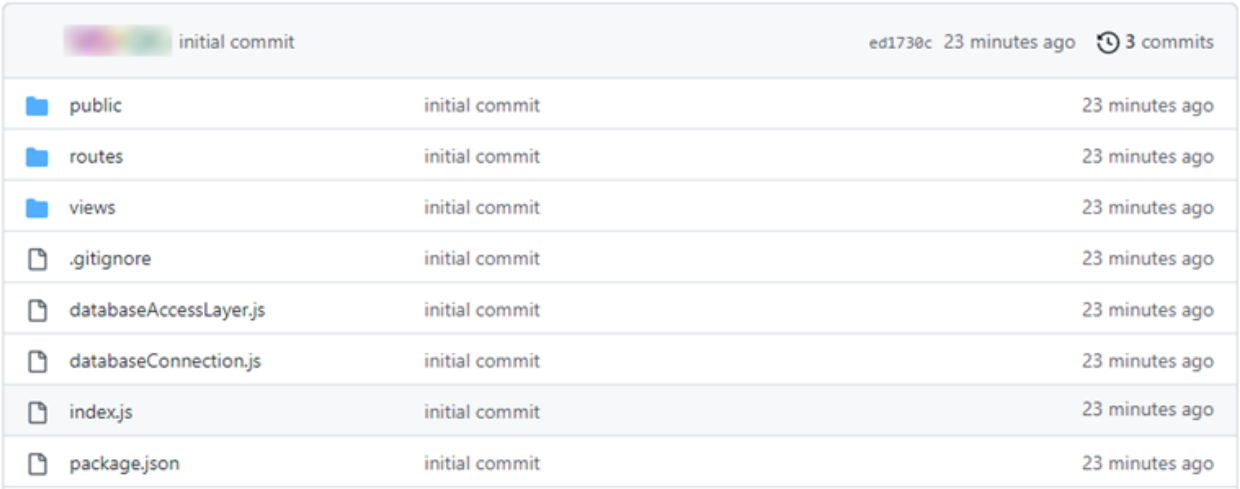
Create a GitHub Repository with the contents of the files within this new folder.
Give your GitHub Repo a name with the following format:
`comp2350-week11-cloudinary-` and your "A0" student number.

Example: `comp2350-week11-cloudinary-A87654321`

Create an initial commit with all the files in the folder.

Push your code to a private repo on GitHub.

Your repo should look something like this on `https://github.com` (when accessed from your account).



Step 3: Create a new Hosted app.

Give your new app a new name according to the same naming convention as your Hosted app: `comp2350-week11-cloudinary-` and your "A0" student number.

Example: `comp2350-week11-cloudinary-A87654321`

Link your new app to your new GitHub Repo.

Step 4: Connect your Hosted app to your MongoDB Atlas.

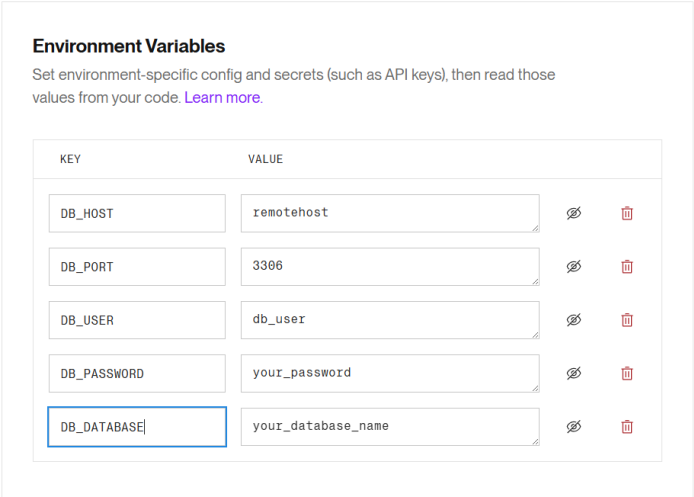
Make a copy of `sample.env` and rename it to `.env` and populate it with the appropriate values.

Remember your MongoDB Atlas connection details (host, port, user, password and database)?

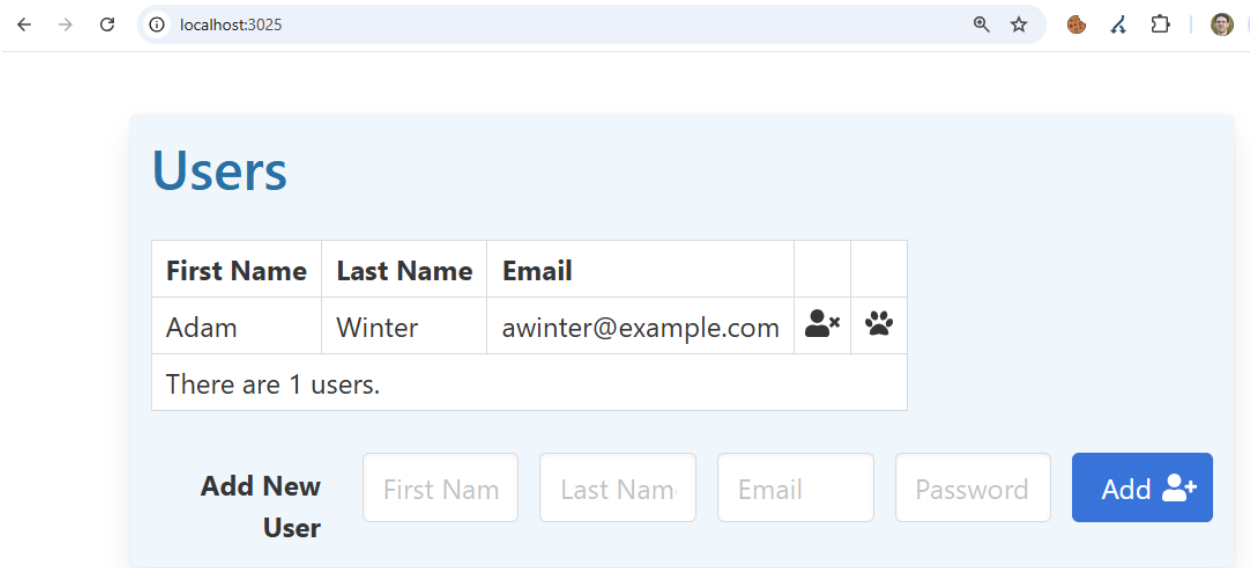
```
REMOTE_MONGODB_HOST=
REMOTE_MONGODB_USER=
REMOTE_MONGODB_PASSWORD=
REMOTE_MONGODB_DATABASE=

CLOUDINARY_CLOUD_NAME=
CLOUDINARY_CLOUD_KEY=
CLOUDINARY_CLOUD_SECRET=
```

IMPORTANT! Remember to add the environment variables for your database connection to your hosted site too!



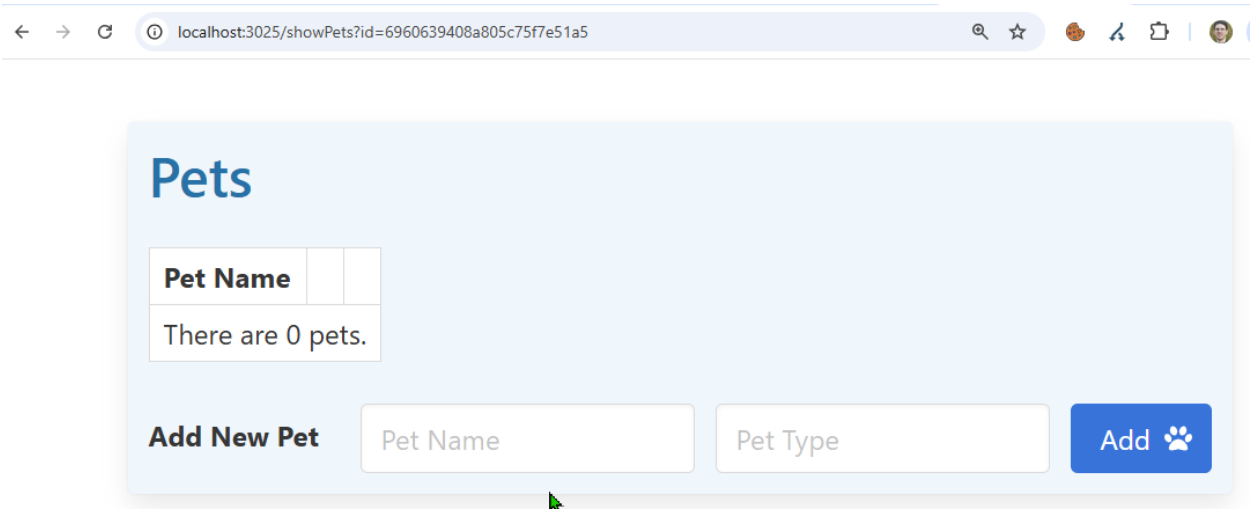
The site should look like this:



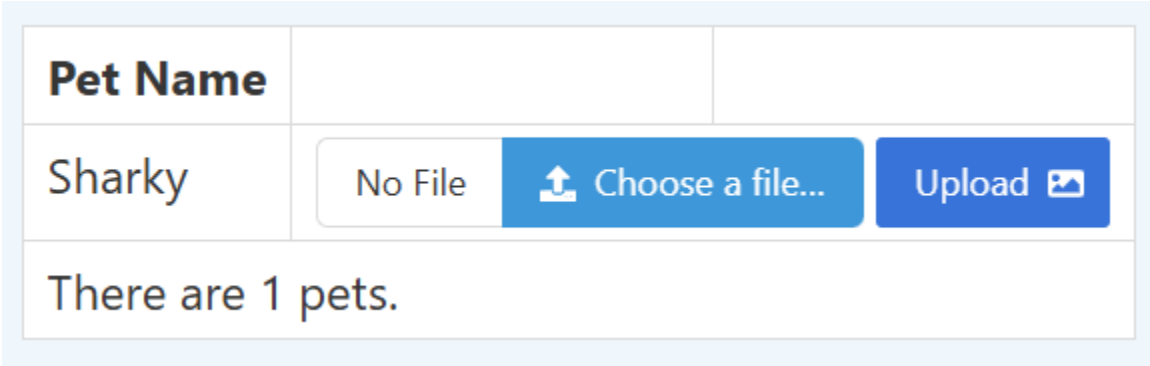
(although your site will probably not have any users yet).

Go ahead and a create a new user (make sure the password is at least 8 characters long).

After you have at least one user, if you click on the paw icon you should get a page like this:

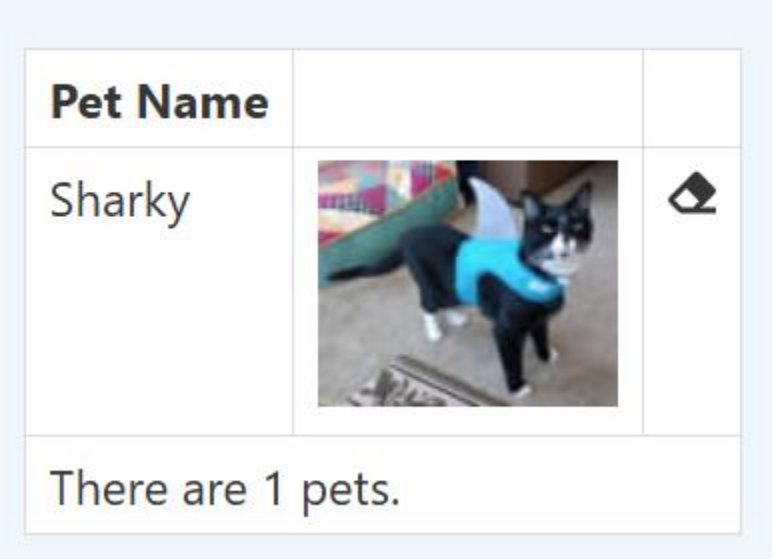


Go ahead and create a new pet.



Once a new pet is there, add an image by clicking "Choose a file…" and then "Upload".

If everything is set up properly, you should be able to upload and view the pet's new image:



Step 5: Let users add images too.

We have seen how the site lets pets have images for them.

GOAL: Your job now is to add the ability for users to add images for themselves too!

Examine the code and the database to learn how the pet images work. Modify the code to allow users to add images to them too.

A few notes:

Pets are linked to users using their object IDs:

```
_id: ObjectId('6960639408a805c75f7e51a5')
first_name : "Maria"
last_name : "Hermanos"
email : "Hermam@example.com"
password : "$2b$12$gtZdK/41cq625SpX056/MeGJSAseOE7fFxyityh2yRwZRvsE90X6."
```

6960639408a805c75f7e51a5

Maria's ObjectID: `6960639408a805c75f7e51a5`

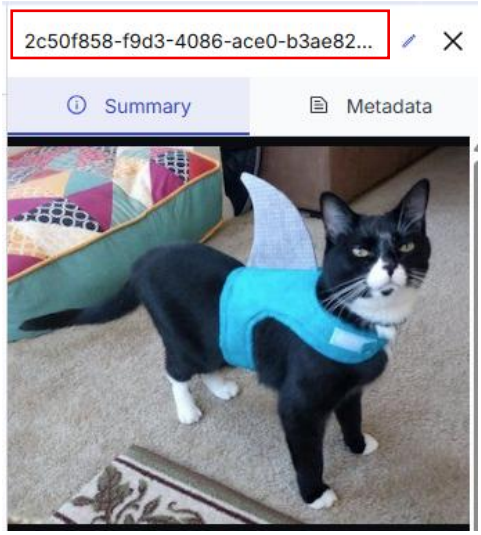
And it is linked to her pets:

```
_id: ObjectId('696063c408a805c75f7e51a6')
name : "Sharky"
user_id : ObjectId('6960639408a805c75f7e51a5')
pet_type : "cat"
image_id : "2c50f858-f9d3-4086-ace0-b3ae824c5a14"
```

(Notice the `user_id` for "Sharky" is the Maria's Object ID: `6960639408a805c75f7e51a5`)

Also, notice how the `image_id`: `2c50f858-f9d3-4086-ace0-b3ae824c5a14`

Matches the `image id` found in Cloudinary:



Submit a text file with the URL of your Hosted App.
Ex: `http://perfect-pet-palace.onrender.com/`

Submit

Filename: **01_Hosted_Cloudinary_Site.txt**

Submit a zip file with your code for your Hosted App.
(excluding the `node_modules/` folder)

Submit

Filename: **02_Hosted_Cloudinary_Code.zip**
The contents of the zip file should contain the following files:

- public
- routes
- views
- databaseConnectionMongoDB.js
- index.js
- package.json
- ReadMe.txt
- sample.env

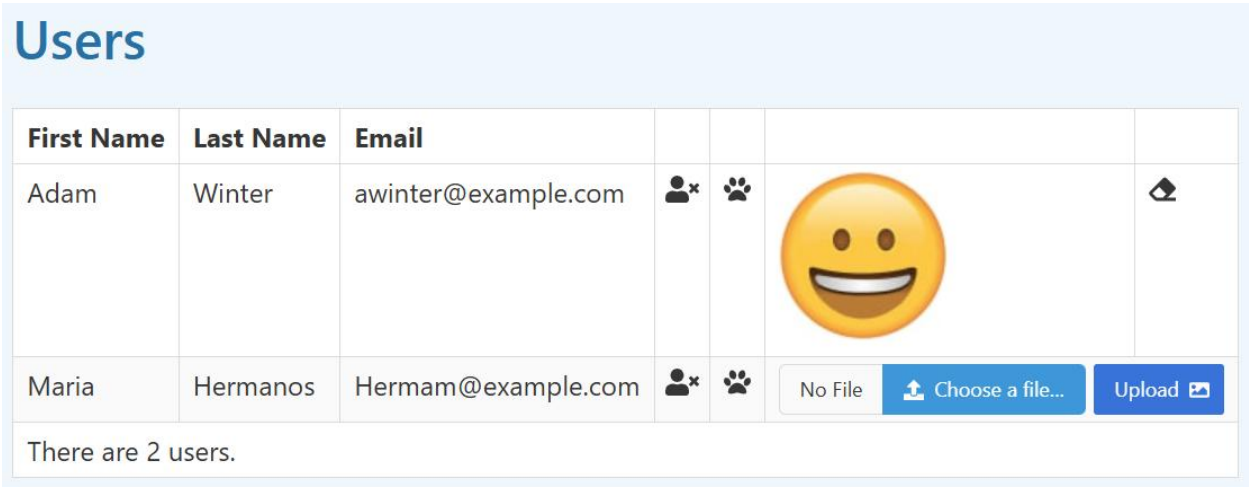
IMPORTANT!! Make sure you **exclude** the `node_modules/` and `.git/` folders!

The `node_modules/` folder can get very large (over 70M) and can easily be re-downloaded using `npm install`.

Submit a screenshot of your Hosted App
with some users added to the database.
Show at least one user with an image.

Submit

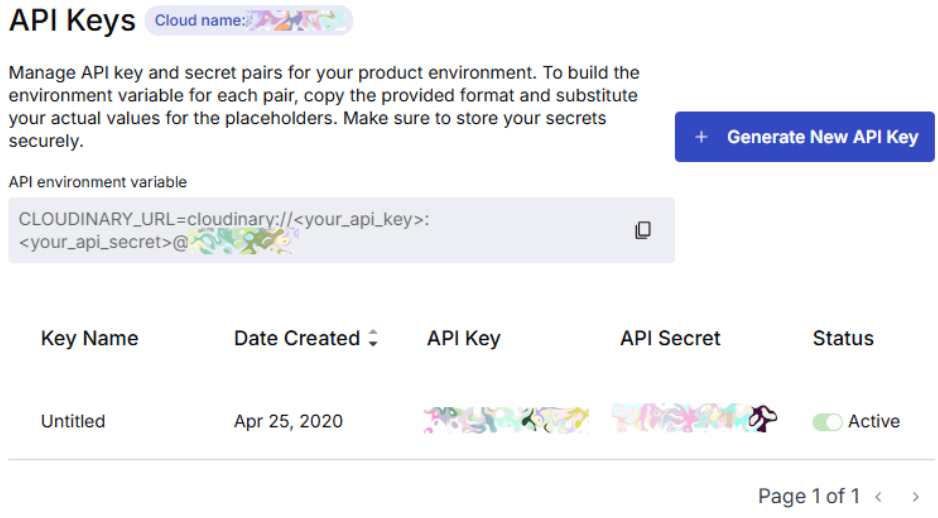
Filename: **03_Hosted_Cloudinary_Screenshot.jpg**



Submit a screenshot of your Cloudinary Settings
From your Cloudinary login, show your API Key, Secret
and Cloud name.

Submit

Filename: **04_Hosted_Cloudinary_Settings.jpg**



Great!